Oxidation-Budgeted Design of Lead-Free $RbGeI_3$ Solar Cells: Back-Surface-Field Enhancement, Realistic Loss Budgeting and SHAP Sensitivity From SCAPS-1D Optimization to Manufacturing Tolerances, Benchmarked Against $CsGeI_3$

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Abstract

The rapid rise of lead-halide perovskite solar cells has been tempered by lead toxicity, motivating a search for lead-free absorbers with comparable optoelectronics. Rubidium germanium iodide ($RbGeI_3$) offers a direct, tunable bandgap, strong absorption, and improved environmental compatibility. Here we report a SCAPS-1D optimization of an $RbGeI_3$ cell organized around its central liability, Ge^{2+} oxidation. After screening eight ETL–HTL combinations, the FTO/TiO_2/$RbGeI_3$/CuI/Au architecture was carried forward on stability and cost grounds (a D2/NiO cross-check through the same pipeline reaches 28.16%, so this stack is not the efficiency optimum). Eleven parameters were varied sequentially on one frozen deck; the consolidated device reaches 27.06% PCE. A Cu_2O-like back-surface field raises the ideal point to 28.01% and recovers about one point on the identical full-realism stack (20.78% $\rightarrow$ 21.76%). Mapping V_{Ge} vacancy thermodynamics onto the defect ladder converts air exposure into a fab spec ($E^f \gtrsim 0.46$ eV holds the headline at 300 K equilibrium, 0.57–0.65 eV for anneal freeze-in; ambient falls below 20.76%), with a 15.53% current-state rung marking what first fabrication must beat. The solver pipeline is calibration-

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checked against experimental β-$CsPbI_3$ [55], and SHAP analysis recovers bulk defects, bandgap and electron mobility as the dominant levers.
Keywords: lead-free perovskite; $RbGeI_3$; SCAPS-1D; back surface field; oxidation budget; SHAP sensitivity analysis

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I. Introduction

I.A. Background and motivation

Global electricity demand keeps climbing while fossil fuels still supply its bulk with a heavy carbon cost [1, 2]. Solar is the fastest-expanding renewable technology, with installed capacity surpassing 2.2 TW in 2024 and levelized costs still falling [3]; continued growth needs absorbers combining high efficiency with low environmental impact and cost [4]. Devices are commonly classed into three generations — crystalline silicon, thin films, and emerging devices — and halide perovskites lead the third generation with certified single-junction efficiencies above 26% and tandems past 33% [3].

I.B. Perovskite solar cells and the lead problem

Halide perovskites (ABX_3) attract photovoltaics with strong absorption, tunable gaps, ambipolar transport with long diffusion lengths, and weak exciton binding [7, 8], rising from sensitized and meso-superstructured architectures [9, 10]. The champion absorbers $MAPbI_3$ and $FAPbI_3$ hold certified records [3] but contain lead, whose life-cycle contamination motivates Sn, Ge, Bi and Sb substitutes [11, 12]. Tin drew first attention since Sn^{2+} shares the Pb^{2+} lone-pair configuration [11], but Sn^{2+} oxidizes readily to Sn^{4+}, creating deep defects and fast degradation [11, 12]. Germanium offers the same lone-pair structure [13]; among Ge absorbers $RbGeI_3$ stands out for a near-optimal direct gap ($\approx$1.3–1.4 eV), $MAPbI_3$-comparable absorption by first principles [13, 17, 18], and good thermal stability [14].

I.C. Role of numerical simulation and research gap

Because experimental optimization of multilayer devices is slow and expensive, numerical simulation screens materials before fabrication. SCAPS-1D (Ghent University) self-consistently solves Poisson, continuity and drift-diffusion equations [19, 20] and is the workhorse for lead-free design, as a recent review of 54 SCAPS lead-free studies confirms [41] (see their Fig. 1).

The $RbGeI_3$ simulation baseline is Pindolia et al. at 9.77% for FTO/TiO_2/$RbGeI_3$/NiO/Ag [14]; predictions since reach 27.21% [21] and 26.47% [22], each optimizing only a subset of parameters. The TiO_2/CuI combination used here has received comparatively little attention despite CuI's documented advantages as an HTL.

I.D. Contributions and paper organization

We close this gap with a SCAPS-1D study of an FTO/TiO_2/$RbGeI_3$/CuI/Au cell built around one idea: Ge^{2+} oxidation converted into a quantitative fab spec. It contributes: (i) an oxidation defect-tolerance map from V_{Ge} thermodynamics on a frozen-deck eleven-step optimization (27.06% ideal, 20.78% stacked window, D2 like-for-like cross-check at 28.16%); (ii) a Cu_2O BSF variant whose trap-immunity mechanism is isolated by controlled interface attribution (+0.98 pp on the identical realistic stack); and (iii) held-out SHAP design rules plus experimental grounding ($CsPbI_3$ calibration refit, frozen database snapshot as context). Band lineups and primary-source-verified benchmarks support each claim.

II. Literature Review and Research Gap

SCAPS-1D, often combined with DFT, is the workhorse for lead-free design before fabrication. Supplementary Table S1 surveys recent Ge- and Sn-based studies; all values there were cross-checked against primary sources, adopting the J–V-implied efficiency where inconsistent. CuI is a wide-gap ($E_g \approx 3.1$ eV) p-type conductor whose valence maximum aligns with $RbGeI_3$ for hole extraction [24, 25], with single-crystal mobilities above 100 $cm^2\ V^{-1}\ s^{-1}$ and

low-temperature solution processing [26]; yet few SCAPS studies pair it with $RbGeI_3$, and none runs an eleven-parameter optimization on this combination. Comparator structures span TiO_2, PCBM, C_{60}, NiO, CuI, CBTS, Spiro-OMeTAD and Ag/Au contacts; note the Loumachi ETL is C_{60} (not "C6") with an Ag contact [21].

III. Device Structure and Material Selection

III.A. Proposed device architecture

We use a conventional n-i-p planar heterojunction, FTO/TiO_2/$RbGeI_3$/CuI/Au: FTO front electrode, TiO_2 ETL, $RbGeI_3$ absorber, CuI HTL, Au back contact (Figure 1). Material choices are justified below; initial parameters are in Supplementary Table S2.

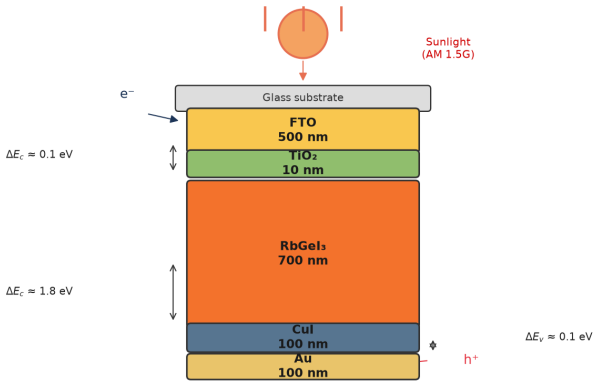


Figure 1: Schematic diagram of the proposed FTO/TiO_2/$RbGeI_3$/CuI/Au perovskite solar cell structure, showing the five functional layers and the direction of photogenerated carrier transport under AM 1.5G illumination.

III.B. Material selection rationale

Fluorine-doped tin oxide (FTO, 500 nm) is the front electrode for its transparency, conductivity, and thermal/chemical stability [27], the standard low-cost TCO choice for this stack. TiO_2 is the ETL: its 3.2 eV gap passes visible light,

its conduction-band minimum gives downhill electron transfer, and its deep valence band blocks holes [28, 29], forming stable perovskite interfaces despite lower mobility than ZnO/SnO_2 [30]. $RbGeI_3$ is the absorber (direct 1.3–1.4 eV gap, $MAPbI_3$-comparable absorption [17, 18], good thermal stability); screening used 400 nm and 1.4 eV from literature [14], both subsequently optimized (Section V). CuI is the HTL (conductive, transparent, 0.10 eV valence offset to $RbGeI_3$, low cost [24, 25, 31]); screening used 100 nm. Au is the back contact, ohmic to CuI [32].

III.C. Initial material parameters

Initial parameters from published experimental and numerical lead-free studies [14, 15, 16, 19, 28] are in Supplementary Table S2 (interfaces: Table S3). All eight configurations were screened at 400 nm absorber (Section V.A, Table II); the consolidated device uses the optimized 700 nm (Section V.G).

IV. Methodology

IV.A. SCAPS-1D simulation framework

The stack was simulated with SCAPS-1D v3.3.10 [19, 20], solving Poisson plus electron/hole continuity self-consistently under drift-diffusion transport (see original references for details).

$$\frac{d}{dx}\left(\epsilon\frac{d\psi}{dx}\right) = -q\left(p - n + N_D - N_A + \rho_t\right) \tag{1}$$

$$\frac{dJ_n}{dx} = q\left(R - G\right) \tag{2}$$

$$\frac{dJ_p}{dx} = q\left(G - R\right) \tag{3}$$

where ϵ is the permittivity, ψ the potential, n/p the carrier concentrations, $N_{\mathrm{D}}/N_{\mathrm{A}}$ the doping, ρ_{t} the trapped charge, J_n/J_p the current densities, and G/R the generation/recombination rates. SCAPS-1D computes the J–V characteristics from the layer properties below.

IV.B. Simulation conditions

All simulations were performed under the AM 1.5G spectrum at an incident power density of 100 mW/cm^2 (one sun, STC) with the operating temperature fixed at 300 K, using the initial material parameters of Supplementary Table S2 and the interface defect densities of Supplementary Table S3 unless otherwise stated. During each optimization step, only one parameter was varied while all remaining parameters were held constant; the general simulation conditions are summarized in Supplementary Table S4. Absorption uses the SCAPS square-root law $\alpha = \mathrm{A}\sqrt{h\nu - E_\mathrm{g}}$ with A = 10^5 cm^{-1} at the electrical bandgap, with radiative recombination at the literature perovskite coefficient (K_rad 10^{-10} cm^3/s) included in all headline runs. Contacts are ideal ($R_\mathrm{s} = 0$, $R_\mathrm{sh} = \infty$) unless stated, so fill factors are upper bounds. All eleven sweeps ran sequentially on one frozen deck generation with carry-forward (Section V.B); failed runs were retried and the survivors surface as `FAILED` rather than stale data. Scripting limits (front-reflection handling, C–V capture, resistance post-corrections) and the full reproducibility protocol are documented in Supplementary Section S7.

IV.C. Performance parameters

Performance was evaluated using the four key parameters extracted from the simulated J–V curve: open-circuit voltage ($\mathrm{V_{OC}}$), short-circuit current density ($\mathrm{J_{SC}}$), fill factor (FF), and power conversion efficiency (PCE), computed as

$$\mathrm{PCE} = (V_\mathrm{OC} \times J_\mathrm{SC} \times \mathrm{FF})/P_\mathrm{in} \tag{4}$$

with P_in = 100 mW/cm^2 under AM 1.5G.

IV.D. Sequential optimization procedure

Starting from the initial device, one parameter at a time (OPAT) was varied while all others were held constant, as detailed in Section V. The procedure comprises eleven optimization steps: (1) absorber thickness (0.3–1.0 μm); (2) absorber bulk defect density (10^{12}–10^{18} cm^{-3}); (3) absorber dielectric constant (10–35, extended until the trend was mapped); (4) absorber electron affinity

(3.7–4.2 eV); (5) TiO_2/$RbGeI_3$ interfacial defect density (10^{12}–10^{18} cm^{-2}); (6) $RbGeI_3$/CuI interfacial defect density (10^{12}–10^{18} cm^{-2}); (7) absorber bandgap (1.2–1.7 eV, with 1.2 probed); (8) TiO_2 ETL thickness (0.01–0.09 μm); (9) CuI HTL thickness (0.05–1.9 μm); (10) conduction-band effective density of states N_C (1×10^{18}–1×10^{20} cm^{-3}, literature floor); (11) valence-band effective density of states N_V (same range). Each step's value was carried forward under a triple-gate adoption rule: sweep maximum, literature-parameter compliance, and Shockley–Queisser admissibility with margin. Parameters that cannot be ordered from a supplier (dielectric constant) or tuned below physical floors (defect densities, N_C/N_V) were pinned at literature values with the full trace disclosed, rather than carried at grid edges; the consolidated set is summarized in Table IV. Because each per-step value is identified with all other parameters fixed at prior-step values, the consolidated set may not be a global optimum. To test for missed cross-interactions, Section V.G reports a genetic-algorithm joint search over the six most sensitive parameters, which returns a best-found joint point (not a proven global maximum).

IV.E. Device configurations screened

Eight planar configurations combining two ETLs (TiO_2, PCBM) with four HTLs (NiO, CuI, CBTS, Spiro-OMeTAD) around the $RbGeI_3$ absorber (D1–D8, Table II) were simulated at the initial absorber thickness of 400 nm (Supplementary Table S2) to permit a fair architecture comparison before the thickness optimization of Section V.B.

V. Results and Discussion

V.A. Screening of device configurations

Table II compiles the photovoltaics of all eight configurations: two ETLs (TiO_2, PCBM) crossed with four HTLs (NiO, CuI, CBTS, Spiro-OMeTAD) on the common $RbGeI_3$ absorber. The D4 architecture (TiO_2/CuI) combines well-matched band alignment with an all-inorganic transport stack, avoiding the stability limitations of organic HTLs.

Device	Architecture
D1	FTO / PCBM / $RbGeI_3$ / NiO / Au
D2	FTO / TiO_2 / $RbGeI_3$ / NiO / Au
D3	FTO / PCBM / $RbGeI_3$ / CuI / Au
D4	FTO / TiO_2 / $RbGeI_3$ / CuI / Au
D5	FTO / PCBM / $RbGeI_3$ / CBTS / Au
D6	FTO / TiO_2 / $RbGeI_3$ / CBTS / Au
D7	FTO / PCBM / $RbGeI_3$ / Spiro-OMeTAD / Au
D8	FTO / TiO_2 / $RbGeI_3$ / Spiro-OMeTAD / Au

Table I: Eight device configurations screened in this study. All configurations were simulated at an absorber thickness of 400 nm.

The best FF belongs to D7 (85.01%) and the best PCE to D2 (21.10%), not D4. Because D2 led at screening, it was run through the identical eleven-step pipeline on its own NiO deck (literature-standard NiO parameters, documented in S7; same triple-gate adoptions) as a like-for-like check: D2 carries 0.8 μm / 10^{12} defects / ϵ_r 23.1 / χ 3.9 / 10 nm ETL / 10^{18} densities to a raw 32.87% chain end, and its gate-adopted device reaches 28.16% (1.11 V, 30.32 mA/cm^2, 83.57%) — the screening gap survives optimization essentially intact (+1.32 pp at screening, +1.10 pp adopted), so D4 is not the efficiency optimum of this family. D4 is retained on stability and cost grounds stated plainly: CuI's documented stability and solution-processability advantages over Spiro-OMeTAD [24, 25] plus NiO's moisture sensitivity against D2, and an all-inorganic transport pair either way. Per-step D2 nodes are tabulated in Supplementary Section S1. Screening ran once per configuration on the pre-rebase deck, so Table II is an architecture-ranking record, not headline numbers; the selected D4 stack was fully re-optimized on the rebased frozen deck. Band-alignment conclusions (offsets, extraction paths) transfer across the rebase while absolute PCEs do not.

Dev.	Architecture	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)
D1	FTO/PCBM/ $RbGeI_3$/NiO/Au	0.81	30.10	83.68	20.40
D2	FTO/ TiO_2/$RbGeI_3$/NiO/ Au	0.82	30.46	84.48	21.10
D3	FTO/PCBM/ $RbGeI_3$/CuI/Au	0.81	30.10	83.96	20.47
D4	FTO/ TiO_2/$RbGeI_3$/CuI/ Au	0.81	30.46	79.98	19.78
D5	FTO/PCBM/ $RbGeI_3$/CBTS/ Au	0.82	30.20	82.36	20.40
D6	FTO/ TiO_2/$RbGeI_3$/CBTS/ Au	0.82	30.55	82.13	20.57
D7	FTO/PCBM/ $RbGeI_3$/Spiro/Au	0.81	30.07	85.01	20.71
D8	FTO/ TiO_2/$RbGeI_3$/Spiro/Au	0.82	30.46	83.95	20.97

Table II: Initial photovoltaic performance of the eight screened device configurations, simulated at an absorber thickness of 400 nm. Selected as the baseline for full parametric optimization.

V.B. Optimization summary

Eleven parameters were optimized one at a time on the screened FTO/TiO_2/$RbGeI_3$/CuI/Au stack (absorber thickness, bulk and interface defect densities, dielectric constant, electron affinity, bandgap, transport-layer thicknesses, conduction- and valence-band densities of states); each value was carried forward under the triple-gate rule, giving the consolidated set in Table IV. Full per-parameter traces (Figs. S1–S10) are given in Supplementary Section S1, and the illumination and dark characteristics (Figs. S11–S12) in Section S2. The defect-density step itself is retained below (Section V.C) together with its energy/asymmetry extension, since Ge-related degradation is central to this work.

Step	Sweep range	Adopted	Gate basis
1 thickness	0.3–1.0 μm	0.7 μm	sweep max, plateau
2 bulk N_t	1e12–1e18 cm^{-3}	1e14	passivated-film pin (max at 1e12 shown)
3 ϵ_r	10–35	15	sweep-flat pin
4 χ	3.7–4.2 eV	3.9 eV	interior max (3.7 non-convergent)
5/6 interface N_t	1e12–1e18 cm^{-2}	1e14	passivated-film pin (max at 1e12 shown)
7 E_g	1.2–1.7 eV	1.4 eV	SQ-margin gate (max at 1.3 rejected)
8 TiO_2	10–90 nm	10 nm	sweep max + pinhole caveat
9 CuI	0.05–1.9 μm	100 nm	sweep-flat pin
10/11 N_C/N_V	1e18–1e20 cm^{-3}	1e18	literature floor (sub-floor $\leq$0.13 pp)

Table III: Per-step sweeps, adopted values and gate basis. Full traces: Figs. S1–S10.

V.C. Effect of absorber defect density

Swept 10^{12}–10^{18} cm^{-3}, PCE peaks at 25.27% at the 10^{12} cm^{-3} floor, reads 23.37% at the adopted passivated-film pin of 10^{14} cm^{-3}, then collapses (23.37 $\rightarrow$ 6.79%) through falling V_{OC} (0.964 $\rightarrow$ 0.559 V), saturating past 10^{17} cm^{-3}. Beyond the total trap density, the energetic position E_t of the dominant defect and the electron/hole capture asymmetry (σ_n, σ_p) were swept to model Ge-related degradation explicitly. At the H1 baseline ($N_t = 10^{14}$ cm^{-3}), moving E_t from 0.2 eV to 1.2 eV above E_v traces a shallow symmetric minimum centred at mid-gap: PCE falls from 27.27% at the band edges to 27.06% for E_t = 0.4-1.0

eV (Fig. 10), the textbook SRH fingerprint that mid-gap states are the most lethal recombination centres. The level position is thus immaterial at baseline densities (27.06-27.27%). A two-point check at $N_t = 10^{15}$ cm^{-3} gives 25.07% (0.2 eV) and 24.40% (0.7 eV), consistent with the bulk-ladder penalty rather than a level effect. Capture asymmetry at mid-gap (Fig. 11) is significant: symmetric large traps (σ_n=σ_p=10^{-17} m^2) collapse to 20.76%, the worst case; at fixed large σ_n=10^{-17} m^2, shrinking σ_p recovers 21.91% (10^{-19} m^2) and then 27.59% (10^{-21} m^2); small σ_n ($\leq 10^{-19}$ m^2) is benign at any σ_p (26.99–27.80%). Electron capture sets the damage. The oxidation threat ($Ge^{2+} \rightarrow Ge^{4+}$, Ge vacancies) therefore cannot be reduced to a single N_t number: its energy and capture character jointly set the damage, and dense symmetric traps are the worst case.

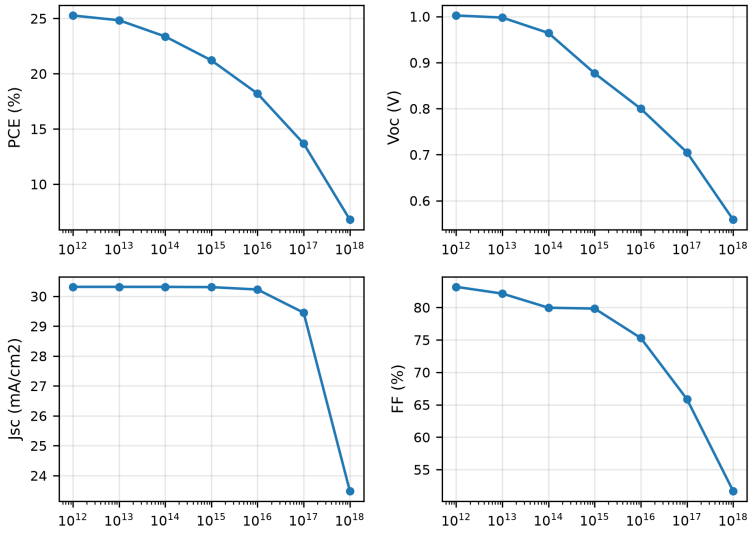


Figure 2: Variation of PCE, V_{OC}, J_{SC}, and FF with $RbGeI_3$ absorber defect density. Performance degrades sharply above 10^{14} cm^{-3}.

V.D. Energy band diagram analysis

The band diagram under illumination (Figure 3) shows clean extraction at both heterojunctions. At CuI/$RbGeI_3$ the 1.8 eV conduction offset blocks

electrons while the 0.10 eV valence step passes holes; at $RbGeI_3/TiO_2$ a 0.10 eV conduction step assists extraction while a 1.90 eV valence offset blocks holes. The peak bulk quasi-Fermi splitting reaches ≈1.13 V while the terminal V_{OC} is 1.06 V: the 70 mV gap is interface recombination plus quasi-Fermi bending toward the contacts (visible in Fig. 3), i.e. a quantified contact loss, not an inconsistency.

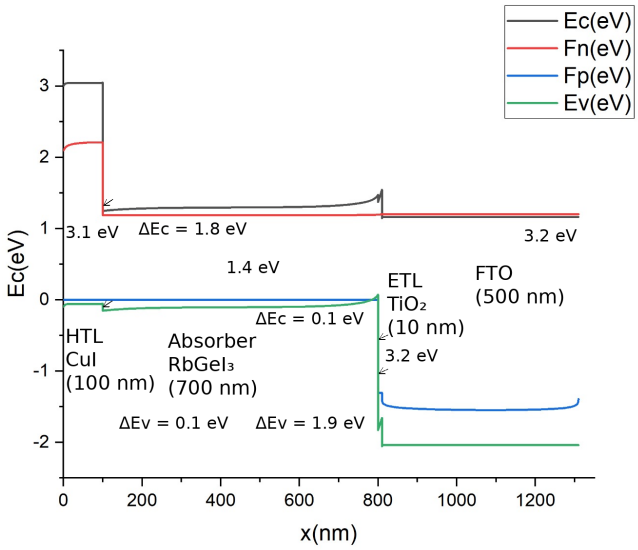


Figure 3: Energy band diagram of the optimized $FTO/TiO_2/RbGeI_3/CuI/Au$ solar cell under illumination, showing the conduction-band edge (EC), valence-band edge (EV), electron quasi-Fermi level (F_n), and hole quasi-Fermi level (F_p). The 1.8 eV conduction-band offset at the $CuI/RbGeI_3$ interface suppresses electron back-transfer, while the 0.10 eV conduction-band offset at the $RbGeI_3/TiO_2$ interface aids electron extraction. The 1.90 eV valence-band offset at the $RbGeI_3/TiO_2$ interface provides an energetic barrier against hole injection into TiO_2.

V.E. Current density–voltage characteristics

The consolidated J–V under AM 1.5G (Figure 4) is sharply rectangular: V_{OC} = 1.06 V, J_{SC} = 30.32 mA/cm^2, FF = 84.11%, PCE = 27.06%, with no s-kink or rollover.

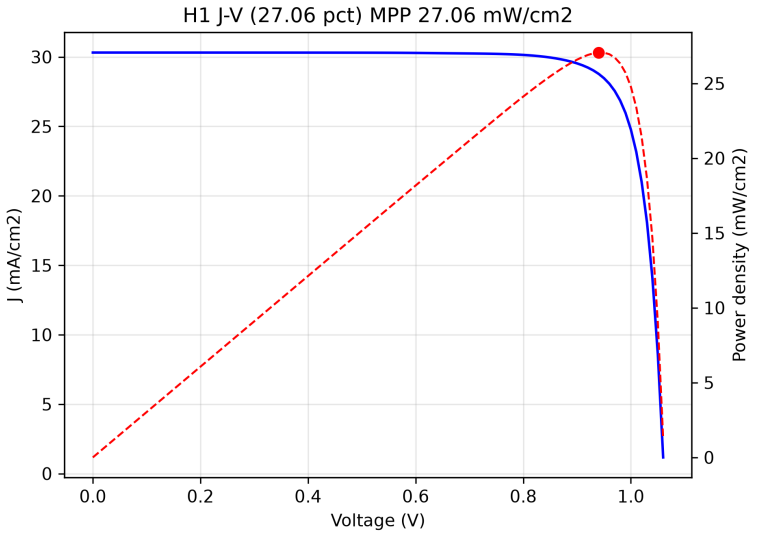


Figure 4: Current density–voltage (J–V) curve of the optimized FTO/TiO_2/$RbGeI_3$/CuI/Au device under AM 1.5G illumination. The red dashed curve shows the output power density, with the maximum-power point marked.

V.F. Summary of optimized device parameters and final performance

After completing the eleven sequential steps, the adopted values were incorporated into the proposed FTO/TiO_2/$RbGeI_3$/CuI/Au cell (Table IV): adopted means triple-gate values (Section IV.D), not raw sweep maxima — defect densities pinned at the passivated-film 10^{14} level, N_C/N_V at the 10^{18} literature floor, dielectric constant at 15 (sweep-flat), CuI at 100 nm (sweep-flat). The consolidated device was simulated in a single SCAPS-1D run with detailed-balance-consistent radiative recombination throughout: V_{OC} = 1.06 V, J_{SC} = 30.32 mA/cm^2, FF = 84.11%, PCE = 27.06%. The V_{OC} climb from the 0.81 V screening point is accounted for: bulk N_t $10^{15}\rightarrow10^{14}$ adds ≈0.09 V (0.877→0.964 V in the sweep), and the N_C/N_V literature floor adds ≈0.09 + 0.06 V across their sweeps, with the remainder from thickness/affinity alignment — no single step carries it. Every metric sits below its Rühle Shockley–Queisser limit at 1.4 eV (32.88 mA/cm^2, 1.122 V, 89.3%, 32.91%) [34]. A six-parameter genetic-algorithm search (population 10, six generations, 62 SCAPS evaluations;

crossover plus Gaussian mutation with elitism; seeded, archived) returns a best-found joint point at E_g = 1.36 eV, $N_t \approx 1.6 \times 10^{13}$ cm^{-3}, interface $N_t \approx 6 \times 10^{11}$ cm^{-2}, $N_C/N_V \approx 1.6 \times 10^{18}$ cm^{-3}, absorber 0.72 μm, giving 1.04 V, 31.55 mA/cm^2, 85.87%, 28.30% (Fig. 7, Table V) — also fully SQ-admissible, but reported as best-found optimizer reach rather than an admissible device: its $N_t \approx 1.6\times10^{13}$ cm^{-3} and interface $N_t \approx 6\times10^{11}$ cm^{-2} sit below the triple-gate floors, not a proven global maximum. Section VI.C stacks all realism losses onto Table IV for a single realistic window (Fig. 9).

Parameter	TiO_2	FTO	$RbGeI_3$	CuI
Thickness (nm)	10	500	700	100
Bandgap E_g (eV)	3.2	3.2	1.4	3.1
Electron affinity χ (eV)	4.0	4.4	3.9	2.1
Dielectric permittivity ϵr	9	9	15	6.5
CB effective DOS N_C (cm^{-3})	2×10^{18}	2.2×10^{18}	1×10^{18}	2.8×10^{19}
VB effective DOS N_V (cm^{-3})	1.8×10^{19}	1.8×10^{19}	1×10^{18}	1×10^{19}
Electron mobility μe (cm^2 V^{-1} s^{-1})	20	20	28.6	100
Hole mobility μp (cm^2 V^{-1} s^{-1})	10	10	27.3	43.9
Donor density N_D (cm^{-3})	9×10^{16}	1×10^{19}	1×10^{9}	0
Acceptor density N_A (cm^{-3})	0	0	1×10^{9}	1×10^{18}
Bulk defect density N_t (cm^{-3})	1×10^{15}	1×10^{14}	1×10^{14}	1×10^{14}

Table IV: Optimized material parameters of the proposed FTO/TiO_2/$RbGeI_3$/CuI/Au perovskite solar cell. Transport-layer bulk defect densities were carried from the initial deck (not among the eleven swept steps).

Note: The optimized interface defect density was 1 × 10^{14} cm^{-2} for both the TiO2/RbGeI3 and RbGeI3/CuI interfaces.

As an arithmetic consistency check, the final-device PCE computed from Eq. (4) using the consolidated J–V metrics is PCE = (1.0613 × 30.3161 × 0.8411) / 100 = 27.06%, matching the SCAPS-1D output to within rounding.

V.G. Uncertainty quantification

Monte Carlo UQ used Latin hypercube sampling (N = 200, seed 11) over eight variables on the H1 deck: gap 1.35–1.45 eV (E_g moves co-move χ), bulk

Device	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)
Adopted reference (Table IV)	1.06	30.32	84.11	27.06
GA best-found (E_g 1.36 eV)	1.04	31.55	85.87	28.30
Reference, N_C=N_V=1e19	0.97	30.32	84.09	24.82
Reference + realistic optics (0.9 sun)	1.06	27.28	83.97	24.28

Table V: Adopted reference device versus GA best-found point and sensitivity rows (defect rows re-simulated, Fig. 8; optics row reports $PCE_{1\text{-sun}}$). PCEs are unrounded SCAPS outputs (displayed inputs rounded).

5×10^{13}–5×10^{15} cm^{-3}, interfaces 5×10^{13}–5×10^{14} cm^{-2} each side, N_C/N_V 5×10^{16}–5×10^{17} cm^{-3}, μ_n/μ_p 15–50 cm^2/Vs.

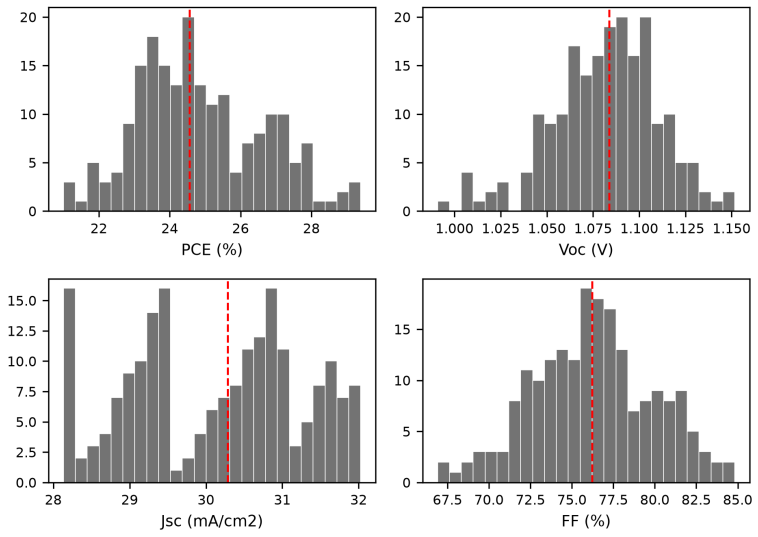


Figure 5: Uncertainty quantification results (N = 200). Histograms of PCE, V_{OC}, J_{SC}, and FF.

Mean PCE 24.86 ± 1.77 % (median 24.55 %; 5th/95th 22.23/27.79 %; 200/200 converged).

V.H. Temperature dependence

Swept 280–400 K (20 K steps) under AM 1.5G.

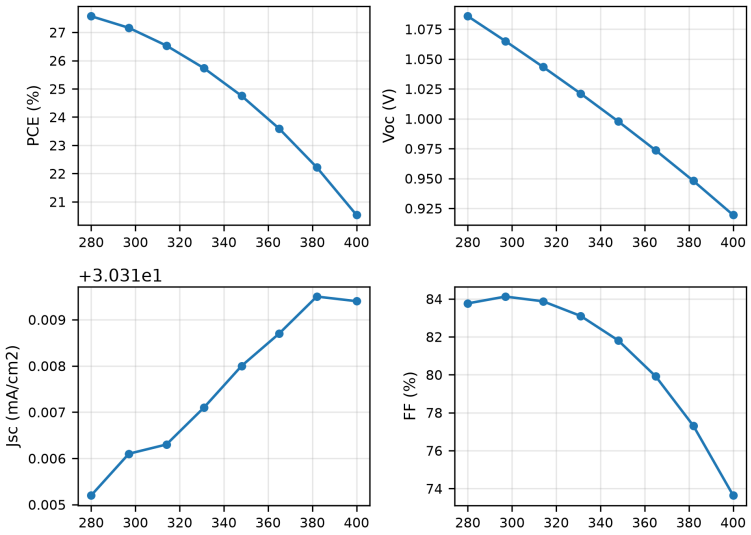


Figure 6: Temperature dependence of photovoltaic parameters from 280 K to 400 K.

$dV_{OC}/dT = -1.38$ mV/K, $dPCE/dT = -0.058$ %/K (PCE falls non-monotonically in FF, so the linear coefficient is descriptive, not predictive); ∼75 % of RT PCE retained at 400 K. $V_{OC}(T)$ extrapolates to $E_a = 1.48$ eV — 0.08 eV *above* E_g, so this is not a clean bulk-recombination signature but an extrapolation artifact consistent with T-dependent N_C/N_V; it is reported as measured, with the artifact disclosed, rather than as proof of bulk-dominated recombination.

V.I. Effect of background acceptor doping

Each $Ge^{2+} \rightarrow Ge^{4+}$ oxidation event donates two holes via V_{Ge}, so background doping is oxidation by another name: sweeping N_A 10^{14}–10^{17} cm^{-3} on the stacked point (Fig. 12), PCE peaks at 23.41% near 3×10^{15} cm^{-3} and softens past 10^{16} cm^{-3} (23.12%) before falling at higher doping (22.17% at 10^{17}) — consistent with the oxidation budget's $p \approx 2[V_{Ge}]$ expectation in Section VI.H, with the penalty soft (≤0.3 pp at 10^{16}), not a collapse. Synthetic target: $N_A \leq 10^{16}$ cm^{-3}. Tying doping to defects per the budget (N_t=10^{15} with N_A=2×10^{15} cm^{-3}) gives 25.52% against 24.40% untied, so the near-intrinsic absorber used

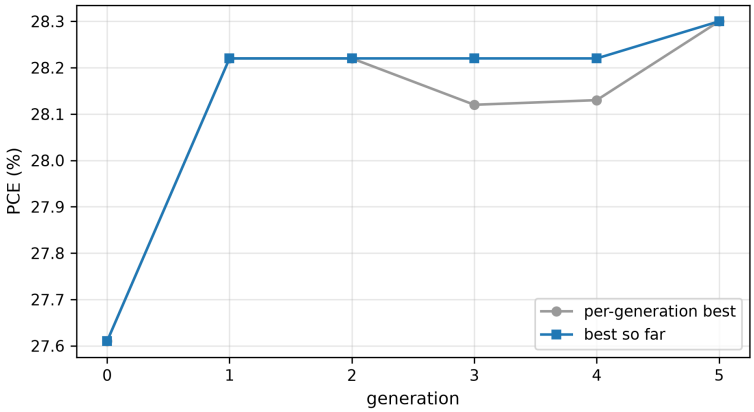


Figure 7: Genetic-algorithm joint search over bandgap, bulk/interface defect densities, N_C, N_V, and absorber thickness. Per-generation best (grey) fluctuates under mutation; best-so-far (squares) reaches 28.30% at generation 5 from 27.61% at generation 0.

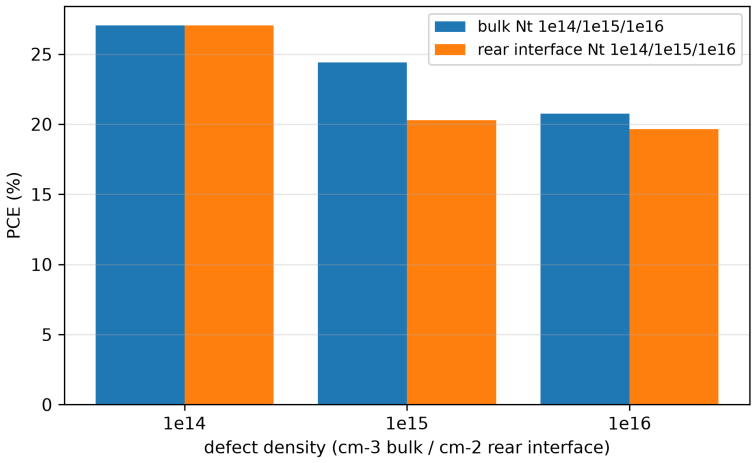


Figure 8: Defect tolerance of the adopted device. Raising bulk N_t to 10^{15} (10^{16}) cm^{-3} drops PCE to 24.40% (20.76%); raising rear-interface N_t to 10^{15} (10^{16}) cm^{-2} drops PCE to 20.29% (19.65%). J_{SC} is nearly unchanged; V_{OC} and FF collapse.

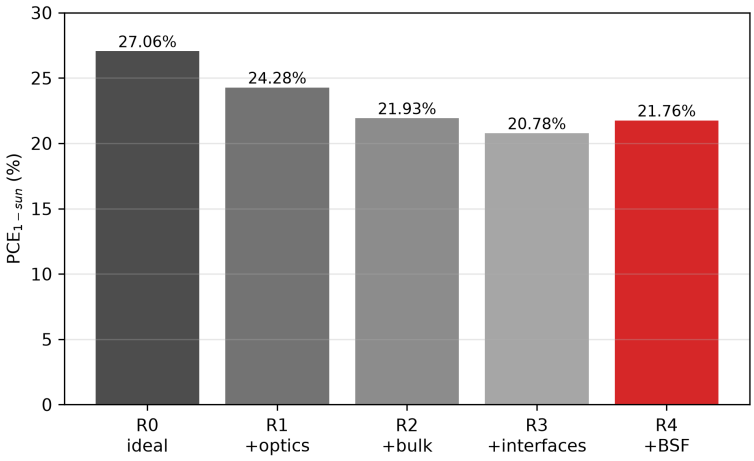


Figure 9: From the adopted 27.06% to the stacked realistic PCE: optics, bulk and interface losses give 20.78% (21.76% with BSF). All nodes re-simulated $PCE_{1\text{-sun}}$ on the adopted deck.

throughout is conservative at this rung, not flattering.

V.J. Effect of series resistance: manufacturing tolerance

Corrections were recomputed on the stacked realistic point (20.78%), not the ideal: FF falls 77.75 → 59.75% and PCE 20.78 → 15.96% as R_s goes 0 → 8 Ω·cm^2 (Fig. 13, external V − J×R_s). Tolerance: $R_s \leq 4$ preserves 20% pre-optics (23.16 → 20.10%); on the full stack the same R_s gives 18.31%. Shunt analysis on the same point: R_{sh} $10^2/5\times10^2/10^3/\infty$ gives 14.26/19.41/20.09/20.71%, so $R_{sh} \geq 10^3$ Ω·cm^2 holds 20%.

V.K. Absorber mobility viability

Electron/hole mobilities were swept jointly from 1 to 100 cm^2/Vs on the stacked point (Fig. 14): PCE is log-linear (18.35% at 1 to 24.32% at 100). Crystallization target: $\mu \geq 15$ cm^2/Vs.

VI. Extended results: loss budget, sensitivity and experimental grounding

VI.A. Benchmarking data (discussion in Section VII)

Literature benchmarking (Table VI) and the $CsGeI_3$ gap control (Table VII, Fig. 16) are discussed in Section VII; the tables and figures are placed here.

Device Structure	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)	Year [Ref.]
FTO/ TiO_2/$RbGeI_3$/NiO/ Ag	0.5311	28.89	63.68	9.77	2022 [14]
FTO/TiO_2/$CsGeI_3$/CuI/ Au	0.667	22.06	73.49	10.80	2022 [23]
ITO/ C_{60}/$RbGeI_3$/CBTS/ Ag	0.99	33.20	82.80	27.21	2024 [21]
FTO/ TiO_2/$RbGeI_3$/PEDOT:PSS/ Au	0.813	30.84	74.19	18.60	2025 [37]
FTO/ TiO_2/$MASnI_3$/CBTS/ Au	1.030	30.24	83.41	25.98	2026 [39]
FTO/ TiO_2/$MASnI_3$/CZTS/Au	0.96	31.66	67.00	20.36	2025 [40]
FTO/ TiO_2/$CsSnI_3$/P3HT/Au	0.972	34.70	78.21	26.38	2024 [38]
FTO/ TiO_2/$RbGeI_3$/Spiro-OMeTAD/Au	1.1813	26.31	85.16	26.47	2026 [22]
FTO/ TiO_2/$RbGeI_3$/CuI/ Au (this work)	1.06	30.32	84.11	27.06	This work
FTO/ TiO_2/$RbGeI_3$/Cu_2O-BSF/CuI/Au (this work)	1.09	30.32	84.62	28.01	This work

Table VI: Comparison of the proposed device with recently reported lead-free perovskite solar cells. All comparator values verified against primary sources. All entries are SCAPS-1D simulation studies.

Device	Condition	V_{OC} (V)	J_{SC} (mA/cm^2)	PCE (%)
$RbGeI_3$	Ideal (N_t=1e14)	1.06	30.32	27.06
$CsGeI_3$	Ideal (N_t=1e14)	1.10	23.18	19.65
$RbGeI_3$	Realistic (N_t=1e15)	1.02	30.31	24.40
$CsGeI_3$	Realistic (N_t=1e15)	1.10	23.17	19.31
$RbGeI_3$	High oxidation (N_t=1e16)	0.96	30.22	20.76
$CsGeI_3$	High oxidation (N_t=1e16)	1.09	23.10	17.07

Table VII: $RbGeI_3$ versus $CsGeI_3$ in the identical FTO/TiO_2/absorber/CuI/Au stack under ideal, realistic, and high-oxidation defect budgets. Only the absorber bandgap differs (1.4 vs 1.6 eV) at fixed affinity — a gap-effect control, not A-site chemistry proof.

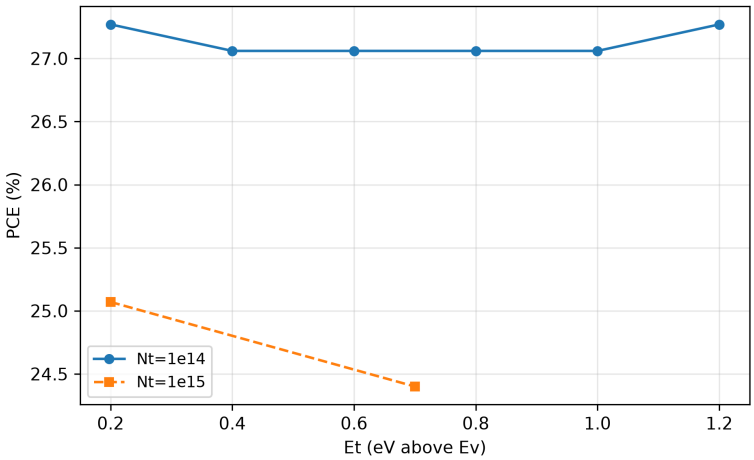


Figure 10: PCE versus defect energy E_t on the H1 baseline ($N_t = 10^{14}$ cm^{-3}). Symmetric minimum at mid-gap (27.06%) recovering to 27.27% at both band edges: the SRH fingerprint.

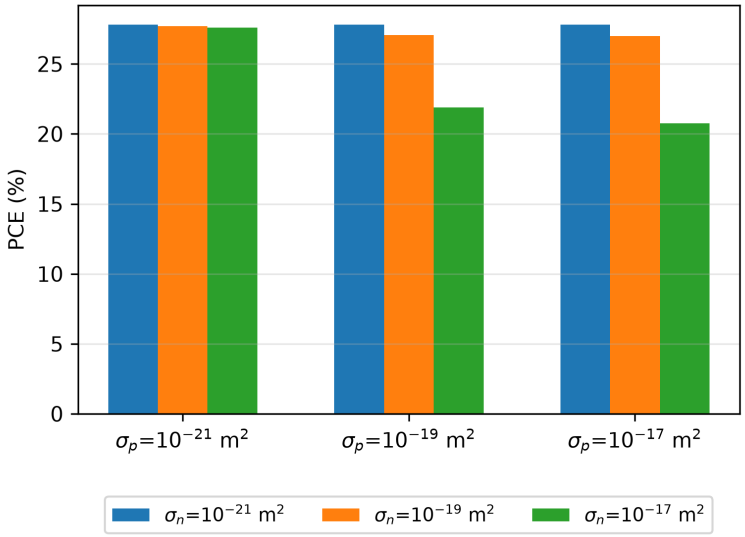


Figure 11: PCE versus capture cross-section asymmetry at mid-gap E_t on the H1 baseline. Symmetric large traps (20.76%) are worst; at σ_n=10^{-17} m^2, shrinking σ_p recovers 21.91% then 27.59%; small σ_n is benign throughout.

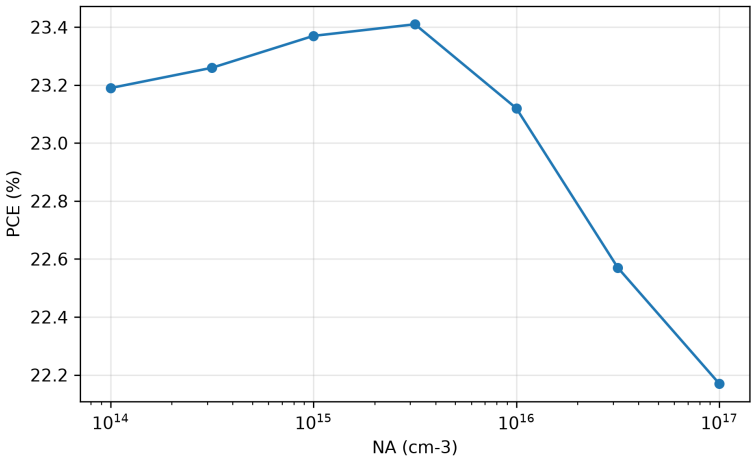


Figure 12: PCE versus background acceptor density on the stacked point. Optimum near 3×10^{15} cm^{-3} (23.41%); penalty soft at 10^{16} cm^{-3} ($\leq$0.3 pp), collapse beyond.

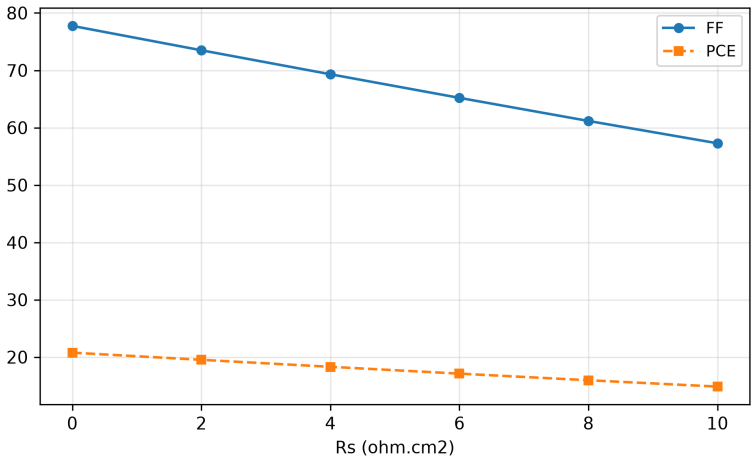


Figure 13: PCE and FF versus series resistance (external V - J x R_s correction) on the stacked 20.78% point. PCE falls to 15.96% at $R_s = 8\ \Omega{\cdot}cm^2$; $R_s \leq 4$ preserves 20% only pre-optics.

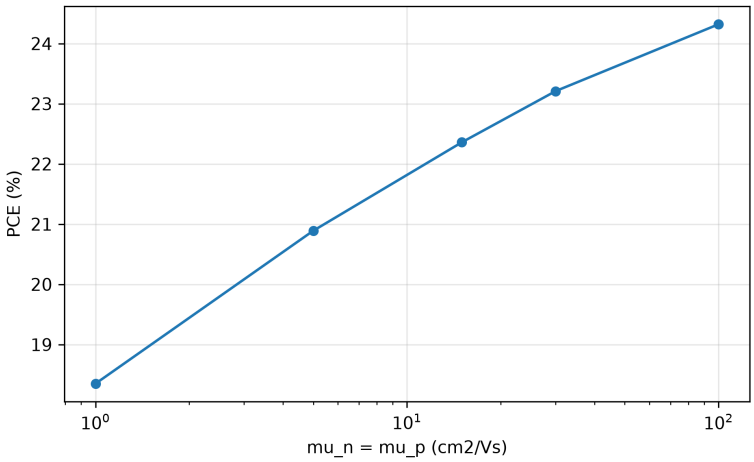


Figure 14: PCE versus absorber mobility ($\mu_n = \mu_p$) on the stacked point. Log-linear rise 18.35% (1) to 24.32% (100 cm^2/Vs); target $\mu >= 15$ cm^2/Vs.

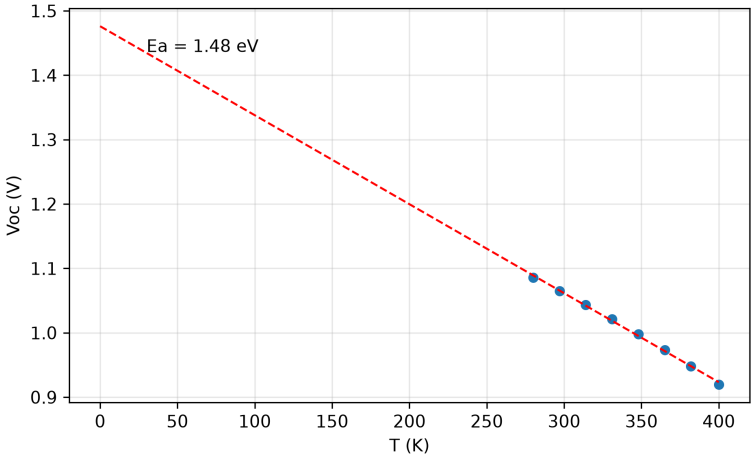


Figure 15: V_{OC} versus temperature with 0 K extrapolation. E_a = 1.48 eV sits 0.08 eV *above* E_g: extrapolation artifact (T-dependent N_C/N_V), not a bulk-recombination signature; dV_{OC}/dT = -1.38 mV/K.

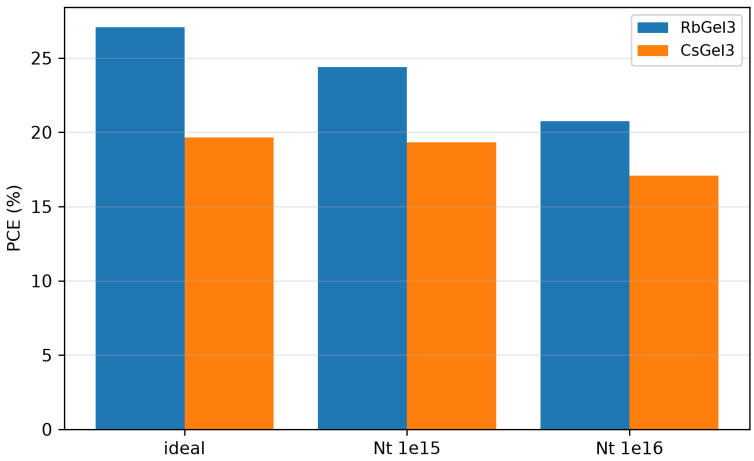


Figure 16: $RbGeI_3$ versus $CsGeI_3$ under ideal, realistic, and high-oxidation defect budgets in the identical stack. $RbGeI_3$ leads on PCE at every level (gap harvest); Cs V_{OC} pinning signals interface limitation.

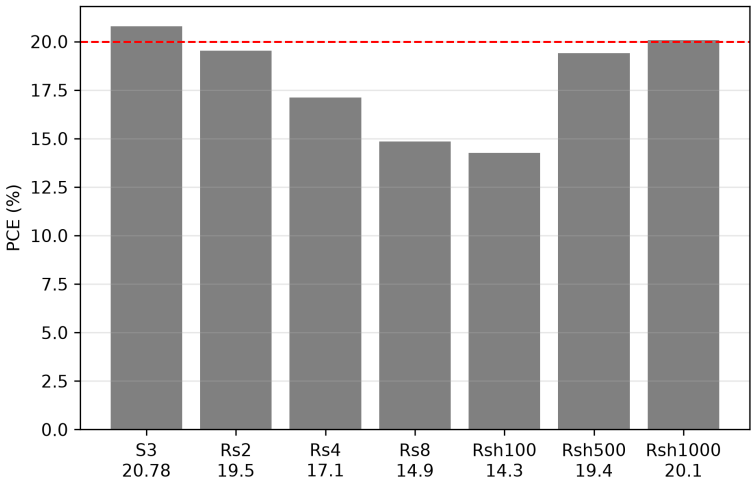


Figure 17: Resistance derating ladder on the stacked 20.78% point (R_s/R_{sh} post-corrections; red dashed line marks 20%).

VI.B. BSF enhancement of the $TiO_2/RbGeI_3/CuI$ stack

A Cu_2O-like p^+ back-surface field (BSF) inserted between $RbGeI_3$ and CuI (E_g 2.1 eV, χ 3.2 eV, ϵ_r 7.5, literature-standard Cu_2O parameters for SCAPS perovskite studies [45]) upgrades the same economical stack without changing the absorber or transport pair. The 50 nm, N_A 10^{18} cm^{-3} pilot converges to 27.93% (+0.87 pp over the 27.06% headline). The full 4×3 grid (Fig. 18; nodes in Supplementary Table S5) shows doping, not thickness, controls the gain: 10^{17} gives 27.55–27.60%, 10^{18} gives 27.93–27.94%, 10^{19} gives 28.00–28.01%. The optimum reaches 28.01% (1.0915 V, 30.32 mA/cm^2, 84.62%); J_{SC} is unchanged, so the gain is V_{OC} (+30 mV) plus FF (+0.5 pp). The mechanism is rear-interface gap widening, not electron reflection (CuI already blocks electrons with 1.8 eV): the BSF removes the 0.10 eV valence cliff – computed from the stated layer offsets, cf. the no-BSF lineup in Fig. 3 – that otherwise shrinks the interface recombination gap, and it halves the rear trap penalty — raising rear N_t to 10^{16} cm^{-2} costs 7.41 pp without BSF (27.06 → 19.65%) but 3.64 pp with it (28.00 → 24.36%, Fig. 18, right panel). Block attribution is explicit: both rear

blocks (CuI/BSF and BSF/$RbGeI_3$) carry the raised N_t, and raising only the far CuI/BSF block costs nothing, pinning the damage on the absorber side. A p^+ control interlayer with CuI bands (no new offset, same N_A) gains only +0.76 pp ideally (27.82%) and still loses 6.49 pp under traps (21.33%), so the field contributes a minority share and the offset removal carries the immunity. Gains are non-additive with the N_t=10^{15} ladder for the same reason: once the rear gap is widened, further defect cuts buy less. All metrics sit below the Rühle limits [33, 34, 35]. Cu_2O is routinely sputtered on perovskite stacks, but a direct Cu_2O-on-Ge-perovskite interface is undemonstrated and flagged as the fabrication risk.

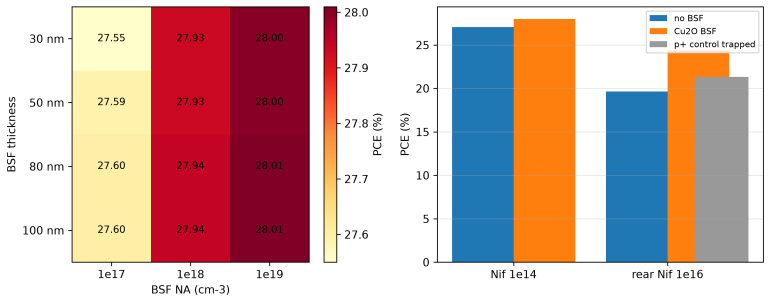


Figure 18: BSF grid PCE (%): N_A $10^{17} \rightarrow 10^{19}$ cm^{-3} rows; thickness 30–100 nm columns. Doping dominates; thickness is flat. Full node values: Supplementary Table S5. Right: rear-interface trap attribution — raising rear N_t to 10^{16} cm^{-2} costs 7.41 pp without BSF (27.06 $\rightarrow$ 19.65%) but 3.64 pp with it (28.00 $\rightarrow$ 24.36%); far-side traps alone cost nothing; the p+ no-offset control (grey) traps at 21.33%.

VI.C. Loss waterfall: what survives realism

Native-defect simulation plus photon-scaling optics and post-processed parasitics bound the experimentally plausible window (Fig. 19; nodes in Supplementary Table S6; $PCE_{1\text{-sun}}$ norm.). Radiative recombination at K_{rad} 10^{-10} cm^3/s costs 0.07 pp (27.13 $\rightarrow$ 27.06%), consistent with the detailed-balance value to order of magnitude at the rebased densities; higher K_{rad} bounds the sensitivity in S6. Auger coefficients 10^{-29}–10^{-27} cm^6/s change nothing at 1-sun ($\pm$0.00 pp) — a reportable null justifying its omission from the headline

budget. Ten percent front reflection costs 2.78 pp (→24.28%); realistic bulk defects (10^{15} cm^{-3}) a further 2.35 pp (→21.93%); realistic interfaces (10^{15} cm^{-2}) a further 1.15 pp (→**20.78%**). The BSF optimum on the *identical* full stack gives **21.76%**, recovering +0.98 pp. The stack is optics- and defect-limited, not transport-limited, and the BSF pays off precisely where reality lives.

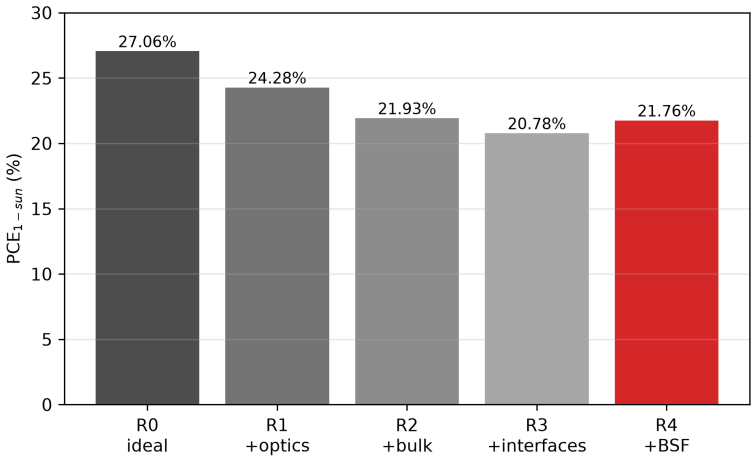


Figure 19: Stacked loss waterfall ($PCE_{1\text{-sun}}$ norm.): optics −2.78 pp, bulk −2.35 pp, interfaces −1.15 pp; BSF on the full stack recovers +0.98 pp. Nodes: Supplementary Table S6.

VI.D. Carrier-lifetime baseline

SRH statistics ($\tau^{-1}=\sigma v_{\mathrm{th}} N_t$) give **1000 ns** at 10^{14} cm^{-3} with diffusion length **8.6 μm** $\gg$ 700 nm absorber, explaining the near-unity collection. Fig. 20 shows the Beer–Lambert generation profile normalized to the H1 J_{SC}. No built-in potential, depletion width or Mott–Schottky values are reported: the near-intrinsic absorber is fully depleted so the depletion approximation does not apply, and C–V extraction is left for fabricated devices (method note in S7).

VI.E. Data-driven global sensitivity analysis

An XGBoost surrogate on the 200-point LHS-UQ block (8 features: E_g, N_t, N_C, N_V, μ_n, μ_p, both interface N_t; 80/20 held-out split, seed 7; 200/200 converged; E_g moves co-move χ so gap physics, not misalignment, is measured)

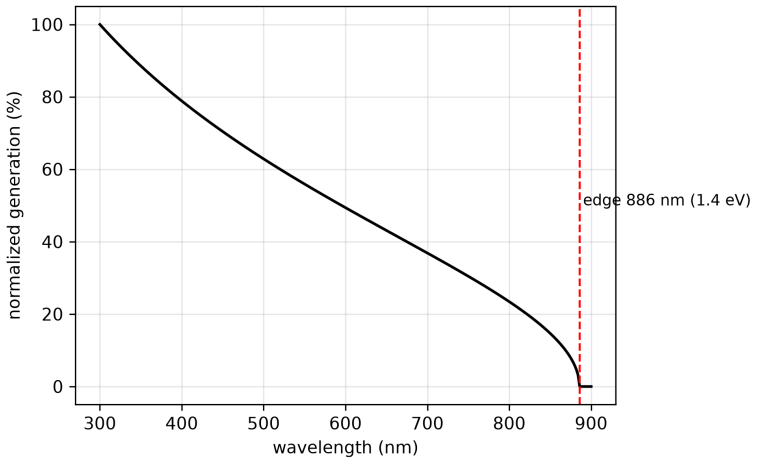


Figure 20: Normalized Beer–Lambert generation profile (square-root law, edge 886 nm). Recombination-profile export is unavailable for this deck (S7); the BSF mechanism rests on the controlled attribution runs (Fig. 18) instead.

gives held-out R^2 0.907, RMSE 0.57 (Fig. 21; random forest agrees at 0.907). SHAP ranks bulk $N_t > E_g > \mu_n >$ CuI-side $N_t >$ TiO_2-side $N_t > N_V > \mu_p > N_C$ (Fig. 22): near the adopted point, bulk defect density dominates, electron mobility outranks hole mobility, and N_C/N_V are inert *within the sampled neighborhood* (5×10^{16}–5×10^{17} cm^{-3}) — consistent with Table V, where 10^{19} costs 2.2 pp outside it. Three hedges: rankings inherit sampled ranges (E_g spans 1.35–1.45 while N_t spans two decades); N_C/N_V are sampled an order of magnitude below the enforced 10^{18} floor, so their "inert" ranking is a neighborhood statement, not a license to lower them; the interface range tops out at 5×10^{14} cm^{-2}, excluding the realistic point. The deterministic-solver R^2 is reported on held-out data only; no factorial cross-check figure is shown (the prior 193-row block ran on the pre-rebase deck).

VI.F. Pipeline calibration against an experimental inorganic perovskite

No $RbGeI_3$ device has been fabricated to date, so direct validation is impossible; we instead run a calibration sanity check of the solver pipeline on the experimentally realized analog β-$CsPbI_3$ of Wang et al. [55] (FTO/c-

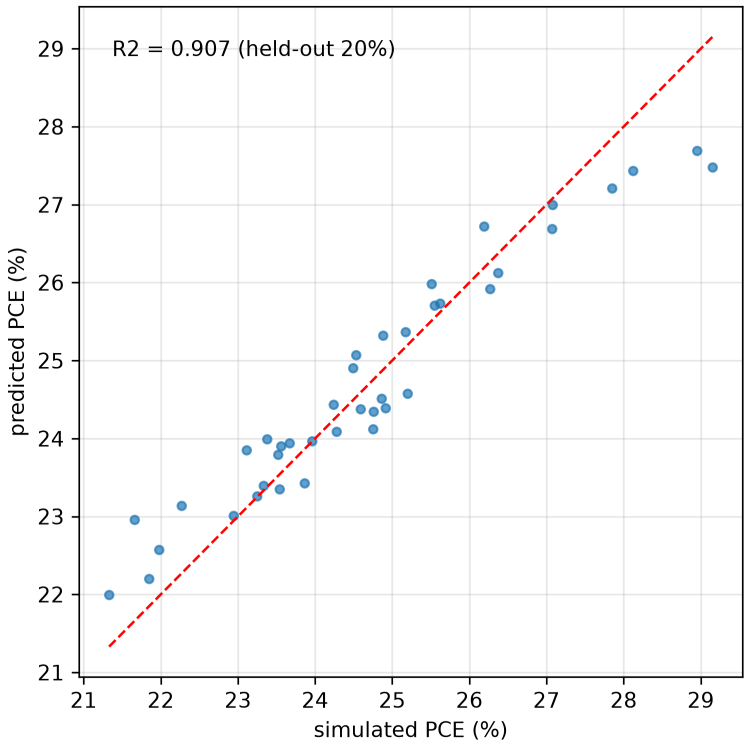


Figure 21: XGBoost parity on the UQ block, held-out 20% (R^2 0.907).

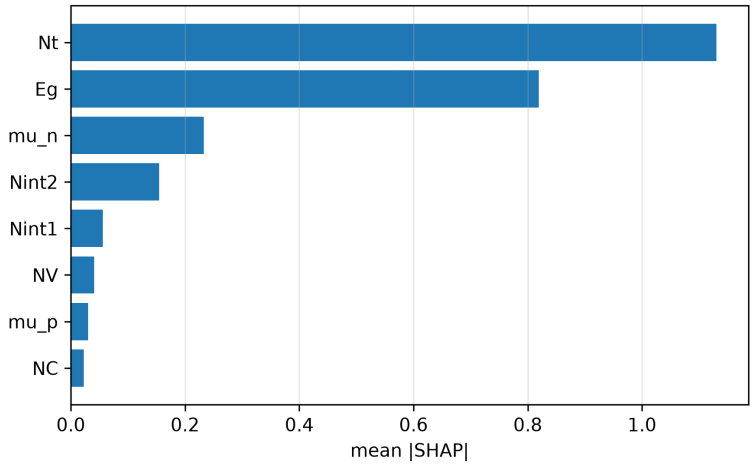


Figure 22: SHAP importance from the UQ surrogate (R^2 0.907): bulk defects dominate, then gap and electron mobility.

TiO_2/$CsPbI_3$/spiro-OMeTAD/Ag, 375 nm absorber, J–V measured at 318 K). Rebuilding that stack with literature-pinned bands and fitting only bulk N_t $\times$ interface N_t reproduces both reported devices (Table VIII, Fig. 23): the CHI-passivated champion at N_t 3×10^{13} cm^{-3} / $N_{t,\mathrm{if}}$ 3×10^{13} cm^{-2} (18.49% vs 18.4%, all metrics within 2%), and the control at N_t 3×10^{13} cm^{-3} / $N_{t,\mathrm{if}}$ 10^{14} cm^{-2} plus an effective R_s 6.9 Ω·cm^2 first-order correction (15.18% vs 15.1%). The control's low FF (0.72) cannot be reached by defects (floor $\sim$0.84) and carries the documented cracks, pinholes and hysteresis, justifying the resistive term; the control fit visibly misses the knee, so it is presented as achieved with that caveat. This calibrates that SCAPS-1D with literature bands can reproduce two measured inorganic devices by tuning defect densities (plus one resistive correction) — it tests the solver pipeline, not CuI, Ge chemistry, the oxidation map or the BSF, none of which appear in the analog; full provenance is in Supplementary Section S5.

Device	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)
Wang control (measured)	1.05	20.03	72.0	15.10
SCAPS rebuild + R_s 6.9	1.060	19.92	71.88	15.18
Wang champion (measured)	1.11	∼20.0	82.0	18.40
SCAPS rebuild	1.127	19.92	82.37	18.49

Table VIII: Calibration check: SCAPS rebuild vs measured β-$CsPbI_3$ devices [55] at 318 K. Fitted knobs are defect densities (plus R_s on the control); the control fit misses the knee.

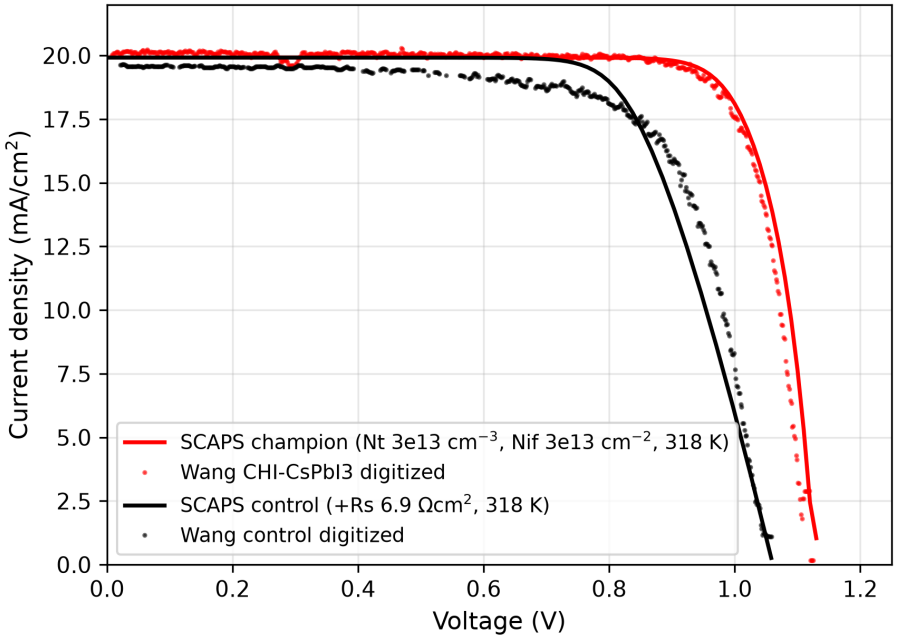


Figure 23: Calibration overlay: SCAPS rebuilds vs digitized J–V of Wang et al. [55] control and CHI-passivated champion (318 K).

VI.G. Inorganic context from a frozen database snapshot

We place the loss-budget window against the Perovskite Database Project via NOMAD [56] as era-bounded context, not a benchmark: 41,316 one-sun entries (43,106 raw fetch; validity filters in S5), of which 2,141 analysis-clean strictly inorganic entries (raw fetch 2,201) include 421 $CsPbI_3$ and just 5 Ge-bearing devices (zero $RbGeI_3$). The snapshot runs to early 2020 plus author uploads, so recent inorganic cells (20.6–21.5%+) postdate it and no superiority over the present state of the art is claimed. Within the snapshot, our stacked 20.78% point (21.76% with BSF) sits above its inorganic max of 18.22% (Fig. 24) — a statement about the gap between simulation upper bounds and that era's inorganic state of the art, not evidence of realism; the refit champion marker (18.49%) is our own calibration point plotted for scale, not an independent device, and its FF (82.37%) sits just above the snapshot's realized FF max (0.82).

VI.H. Oxidation budget: from vacancy thermodynamics to fab spec

Our defect densities are not free parameters: each $Ge^{2+} \rightarrow Ge^{4+}$ oxidation event creates a Ge vacancy (V_{Ge}), the dominant native acceptor in $CsGeI_3$ DFT [47], so air exposure sets N_t directly. Mapping equilibrium V_{Ge} concentrations ($N = N_{\mathrm{sites}} \exp(-E^f/kT)$, $N_{\mathrm{sites}} = 4.87 \times 10^{21}$ cm^{-3}) onto the consolidated N_t ladder (Fig. 25) yields a quantitative fab spec: $E^f \gtrsim 0.46$ eV (good glovebox, Ge-rich chemistry) holds the 27.06% headline; $E^f \approx 0.40$ eV (marginal) lands on the realistic 24.40%; brief air exposure ($E^f \approx 0.35$ eV) pushes $p \approx 2[V_{Ge}]$ past 10^{16} cm^{-3}, where the N_A penalty stays soft ($\leq$0.3 pp); ambient processing ($E^f \approx 0.30$ eV) falls below the high-oxidation 20.76%. These are 300 K equilibrium values and hence lower bounds: defects frozen in from typical 100–150 °C anneals shift every threshold up by 0.11–0.19 eV (holding 10^{14} cm^{-3} then needs $E^f \approx 0.57$–0.65 eV), and the processing-condition bands are illustrative Ef assumptions, not derived Ge/O chemical-potential regimes. At the Ge-poor limit vacancies reach $\sim 10^{19}$ cm^{-3} and the absorber goes metallic, as Ming et al. predict [47] — consistent with rapid Ge oxidation [48] and with no reported

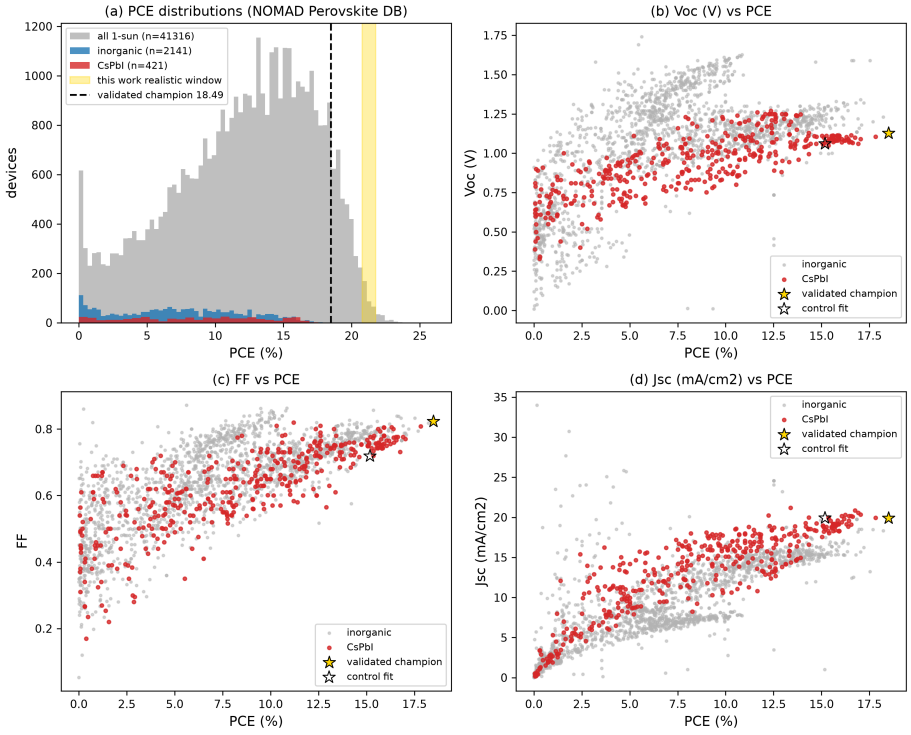


Figure 24: (a) PCE histograms (NOMAD Perovskite DB [56], frozen pre-2021 snapshot): all 1-sun, inorganic subset, $CsPbI_3$; gold band = this work's realistic window; dashed line = calibration refit. (b–d) V_{OC}, FF, J_{SC} vs PCE; gold star = refit champion marker (18.49%, plotted for scale), white star = control fit (15.18%).

$RbGeI_3$ device to date (phase stability of cubic vs. $R3m$ is separately unaddressed here). The mapping is corroborated from three sides: sub-ns TRPL lifetimes invert to $N_t \gtrsim 10^{17}$, so real Ge films today are 100–1000× over budget [54] (S6) — and a current-state rung at N_t=10^{17} cm^{-3} on the adopted stack gives 15.53%, the number a first fabrication should expect to beat; the 1%-Ge optimum in $CsPb(I,Br)_3$ (19.70% [52]) marks the same proliferation threshold from the doping side; and passivation chemistries that remove deep levels — PC-CMAI (+3.29 pp [51]), Lewis bases in DFT [53] — are N_t reduction in waterfall language. The competing V_I deep-trap picture is functional-dependent (benign under HSE+SOC [50]), so this budget rests on V_{Ge} + Cs_I [49]; $CsGeI_3$ energetics transfer to $RbGeI_3$ since the A-site carries no band-edge weight. Derivation, bounds and kinetic caveats in S6.

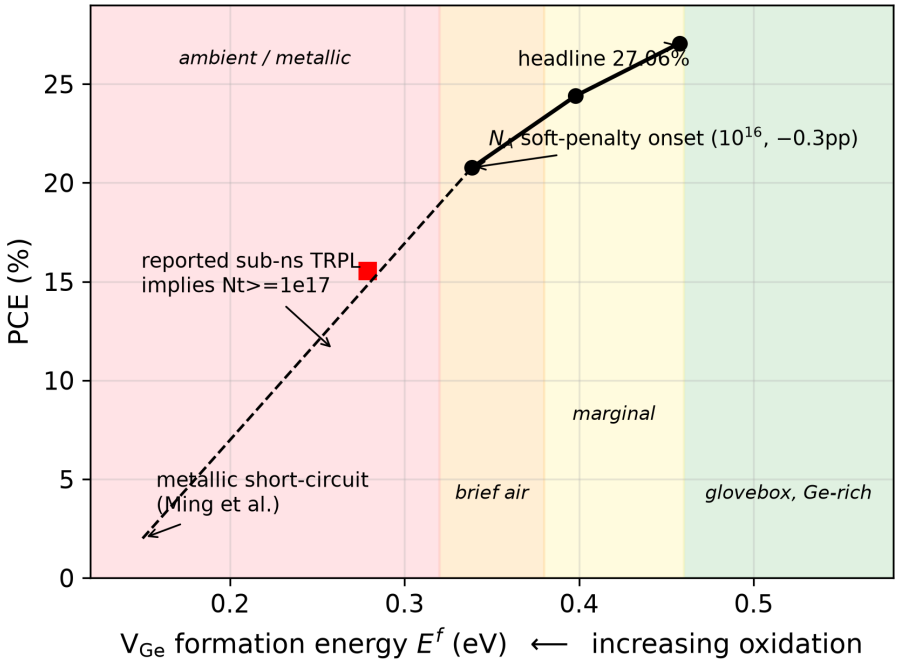


Figure 25: Oxidation budget: consolidated PCE vs V_{Ge} formation energy (dots; equilibrium mapping in text), with processing-condition bands. Dashed extension toward the metallic limit is schematic; TRPL marker from reported sub-ns lifetimes in Ge-containing films (S6).

VII. Comparison with Literature

VII.A. Benchmarking against recent literature

Table VI compares the optimized FTO/TiO_2/$RbGeI_3$/CuI/Au device with recently reported $RbGeI_3$-based and related lead-free Ge/Sn-based perovskite solar cells. The proposed device achieves a PCE of 27.06%, within 0.15 pp of the best reported $RbGeI_3$ value (27.21%, Loumachi et al. [21]), exceeding the 26.47% of Pathak et al. [22] (preprint) by 0.59 pp, the 18.60% of Raj et al. [37] by 8.46 pp, and the 9.77% simulation baseline of Pindolia et al. [14] by 17.29 pp. The performance is competitive with the best tin-based devices (26.38% $CsSnI_3$ [38], 25.98% $MASnI_3$ [39]) while sidestepping $Sn^{2+} \rightarrow Sn^{4+}$ solution oxidation (Ge brings its own $Ge^{2+} \rightarrow Ge^{4+}$ liability, budgeted in Section VI.H), and exceeds CZTS-HTL devices [40]. All comparator values were cross-checked against primary sources (adopting the efficiency implied by reported V_{OC}, J_{SC}, FF where inconsistent); all entries are SCAPS-1D simulations, and the Loumachi string is ITO/C_{60}/$RbGeI_3$/CBTS/Ag (Section II).

In short, the optimized device lands within 0.15 pp of the best published $RbGeI_3$ simulation (27.21% [21]), and stays competitive with the top tin-based lead-free cells, all with an all-inorganic transport stack. With the Cu_2O-like BSF the ideal point reaches 28.01% while keeping the cheap TiO_2/CuI pair (cf. Sb_2S_3-route 30.35% [42] — numerically equal from an independent stack, not the same device — and IGZO/Cu_2O 33.84% [43]); the BSF-realistic 21.76% is the number an experimentalist should target, reported with the full loss decomposition most entries omit. The ML R^2 (0.907 held out) sits with recent SCAPS+ML reports [44, 45, 46]. The Pindolia et al. [14] baseline of 9.77% leaves a wide gap for improvement. Working $RbGeI_3$ solar cells are still rare, so experimental confirmation remains the bottleneck.

VII.B. Gap-plus-alignment control (fixed-χ protocol)

To isolate the gap effect, $CsGeI_3$ was simulated in the identical stack changing only the bandgap (1.4 $\rightarrow$ 1.6 eV) at fixed χ; a control rerun at 1.4 eV

reproduces the $RbGeI_3$ point exactly (27.06%), proving the isolation is computational. The $CsGeI_3$ point (Table VII) gives 1.0966 V, 23.1755 mA/cm^2, 77.3%, 19.65%: Cs wins V_{OC} by 0.035 V through the wider gap while Rb wins J_{SC} by 7.14 mA/cm^2 and FF by 6.8 pp. The J_{SC} gap is dominated by photon harvest — the 1.6 eV gap excludes the entire 775–886 nm band under this protocol — while the 0.30 eV cliff's signature is the pinned V_{OC} ($\sim$1.10 V across a 100$\times$ change in N_t: 19.65/19.31/17.07%) and the FF deficit, i.e. interface limitation rather than intrinsic robustness. This is therefore evidence about gap-plus-alignment, not A-site chemistry proper — stated plainly because the same fixed-χ choice underlies the comparison. The ranking survives derating (10^{15}: 24.40 vs 19.31%; 10^{16}: 20.76 vs 17.07%, Fig. 16).

VIII. Limitations and Future Work

VIII.A. Limitations of the present study

The stacked ladder reads 27.06% ideal $\rightarrow$ 24.28% (optics) $\rightarrow$ 21.93% (bulk 10^{15}) $\rightarrow$ 20.78% (interfaces 10^{15}), with R_s/R_{sh} post-corrections (Fig. 16). Five caveats bound these results. First, the 27.06% adopted device is an ideal-coupling upper bound at literature-pinned densities; the unconstrained sweep maxima (Eg 1.3, N_C/N_V below 10^{18}) were rejected by the triple-gate rule, not hidden. Second, the adopted 10^{14} densities require well-passivated films: bulk 10^{15} (10^{16}) drops PCE to 24.40% (20.76%), rear interfaces 10^{15} (10^{16}) to 20.29% (19.65%), with J_{SC} nearly unchanged and V_{OC}/FF collapsing (Fig. 8); holding 10^{14} in $RbGeI_3$ is hard because Ge^{2+} oxidizes readily to Ge^{4+} in ambient. Third, SCAPS-1D assumes uniform composition and abrupt interfaces that cannot be built exactly. Fourth, the sequential procedure misses cross-interactions by construction; the GA best-found point (28.30%) is reported as optimizer reach, not a proven maximum, and it sits below the triple-gate density floors so it is not an admissible device under our own rules. Fifth, the oxidation map is an equilibrium bound (300 K values are lower bounds; 100–150 °C freeze-in shifts thresholds up 0.11–0.19 eV): real oxidation is kinetic, dose-dependent

and surface-localized, and $CsGeI_3$ energetics transfer to $RbGeI_3$ by framework analogy, not calculation.

VIII.B. Future work

The obvious next step is to build and test the optimized device under strict inert processing (<0.1 ppm O2/H2O glovebox), with Ge^{2+} stabilization (reducing additives, trap passivation, alloying, encapsulation) and stability testing. On the modelling side, three extensions are queued: (i) a full 11-parameter global optimization (Bayesian or surrogate-assisted, ≈500 evaluations) to confirm the six-gene GA maximum; (ii) a transfer-matrix optical model coupled to the drift-diffusion solution using measured $RbGeI_3$ n/k spectra; and (iii) replacing the numerical grading with a realizable composition grade re-optimized in SCAPS.

IX. Conclusion

We optimized a lead-free $RbGeI_3$ perovskite solar cell in SCAPS-1D and converted its central liability — Ge^{2+} oxidation — into a quantitative fab spec. From eight ETL–HTL combinations, FTO/TiO_2/$RbGeI_3$/CuI/Au was selected on durability, cost, and alignment rather than raw efficiency (D2 outscored it 21.10% to 19.78% at screening). After sequential optimization of eleven parameters on one frozen deck, the consolidated device reaches 27.06% (V_{OC} = 1.06 V, J_{SC} = 30.32 mA/cm^2, FF = 84.11%), within 0.15 pp of the best prior $RbGeI_3$ simulation, with every metric below its Shockley–Queisser limit; the D2/NiO cross-check through the same pipeline reaches 28.16%, so the reported stack trades the efficiency crown for stability and cost. Joint search returns 28.30% as optimizer reach below the adoption floors (E_g = 1.36 eV, 62 evaluations), while stacked optical, defect and parasitic losses bound the realistic window to a single number: 20.78% (21.76% with BSF). Defect-energy, doping, resistance, mobility, and gap-control sweeps (Sections V.C, V.J–V.L, VII.B) convert the headline into tolerances recomputed on the stacked point: $R_s \leq 4$ (pre-optics), $R_{sh} \geq 10^3$ Ω·cm^2, $N_A \leq 10^{16}$ cm^{-3}, $\mu \geq 15$ cm^2/Vs. Fixing the economical

TiO_2/$RbGeI_3$/CuI stack, a p^+ BSF adds +0.94 pp ideally (27.06 → 28.00%, 28.01% at the grid optimum) and +0.98 pp on the identical full stack (20.78 → 21.76%) by widening the rear interface gap, with block-level trap attribution and a p^+ control isolating the offset share; and SHAP sensitivity yields design rules (bulk N_t, E_g, μ_n) that one-at-a-time sweeps alone cannot provide. Targets for the lab: N_A(BSF) $\sim 10^{19}$ cm^{-3} ($\geq$30 nm suffices), bulk N_t $\leq 10^{15}$ cm^{-3}, and an oxidation budget of $E^f \gtrsim 0.46$ eV at 300 K equilibrium (0.57–0.65 eV for anneal freeze-in) for the headline, with 15.53% marking the rung current films must beat. The pipeline behind these targets reproduces two experimental β-$CsPbI_3$ devices within 2% in a calibration refit, and its stacked window is placed against a frozen inorganic snapshot as context — the ideal-to-realistic bridge this field needs before first fabrication.

Data availability

All SCAPS decks (initial, H1-adopted, D2, BSF, stacked, attribution controls), run scripts and archived result JSONs listed in Supplementary Section S7 are available from the authors on request; digitized validation curves and database caches are documented in Supplementary Section S5.

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